Crop Recommendation Report

Aurangabad, Maharashtra • 2027

Top Recommended Crop	Sugarcane	Confidence: 22.1%
District	Aurangabad	Maharashtra
Assessment Year	2027	Crops Evaluated: 8

Recommendation Explanation

Sugarcane leads the recommendation for Aurangabad in 2027. It occupies 20.9% of the gross cropped area, reflecting its strategic importance. Combined with an above-average yield of 63466 kg/ha in the previous year, it remains the most suitable crop for the upcoming season.

Full Crop Ranking

Rank	Crop	Confidence (%)	Prev. Yield (kg/ha)	Hist. Mean (kg/ha)	Area Share (%)
1	**Sugarcane**	**22.1%**	**63466**	**67599**	**20.9%**
2	Wheat	16.7%	4531	3772	6.2%
3	Maize	12.0%	3305	3030	3.8%
4	Mustard	10.6%	1279	1340	10.9%
5	Groundnut	10.1%	1521	1571	2.9%
6	Lentil	9.8%	972	972	4.9%
7	Chickpea	9.6%	1110	1172	14.5%
8	Rice	9.2%	3319	3009	35.9%

Key Influencing Factors

Factor	Influence Direction	Magnitude
Monsoon Season Precipitation	→ Neutral	Low
Growing Season Temperature	→ Neutral	Low
Previous Year Yield vs Historical Average	↑ Favourable	High
District Area Share	→ Neutral	Low
Irrigation Coverage Ratio	↑ Favourable	High

Climate Summary

Metric	Value
Annual Precipitation	893 mm
Monsoon Season Precipitation	791 mm
Average Maximum Temperature	31.9 °C

Average Minimum Temperature	17.9 °C
Irrigation Coverage Ratio	80.70%

This report was generated by the AgriCAST AI Crop Recommendation System using a LambdaMART Learning-to-Rank model trained on historical agricultural and climate data. Recommendations are decision-support tools and should be verified with local agronomists before operational use. AgriCAST v1.0